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Experiment No.6 : Throughput and Latency Measurement Using Network Tools

Aim

To measure and analyze **throughput and latency** of a network using tools such as **Ping and Speed Test**.

Objectives

1. To understand the concepts of throughput and latency.
2. To measure network latency using the **Ping** command.
3. To measure download and upload throughput using a **Speed Test**.
4. To compare the obtained results and evaluate network performance.

Theory

1. Throughput:
Throughput is the actual amount of data successfully transferred through a network per unit time.

- Measured in **Mbps (Megabits per second)**.
- Higher throughput generally means better data transfer performance.

2. Latency:
Latency is the time taken for a data packet to travel from the source to the destination and back.

- Measured in **milliseconds (ms)**.
- Lower latency means faster network response.

Example:
If Ping shows **20 ms**, the network has a relatively quick response.
If a speed test shows **50 Mbps download**, data can be downloaded at approximately 50 megabits per second under the test conditions.

Procedure

Part A: Measuring Latency using Ping

1. Connect the computer/mobile to the Internet.
2. Open **Command Prompt** in Windows.
3. Type:

```
ping google.com
```

4. Press **Enter**.
5. Observe the **time in ms** for each reply.
6. Note the minimum, maximum, and average response time.
7. Repeat the test 2–3 times and record the results.

Part B: Measuring Throughput using Speed Test

1. Connect the device to the network.
2. Open a reliable Internet speed-test service.
3. Start the speed test.
4. Wait until the test is completed.
5. Record:
    - **Download speed (Mbps)**
    - **Upload speed (Mbps)**
    - **Ping/Latency (ms)**
6. Repeat the test at least **3 times**.
7. Calculate the average values.

Observation Table

Trial	Ping/Latency (ms)	Download (Mbps)	Upload (Mbps)
1	**34**	**4.68**	**1.90**
2	**30**	**9.18**	**0.20**
3	**72**	**20.43**	**0.24**
Average	**45.33**	**11.43**	**0.78**

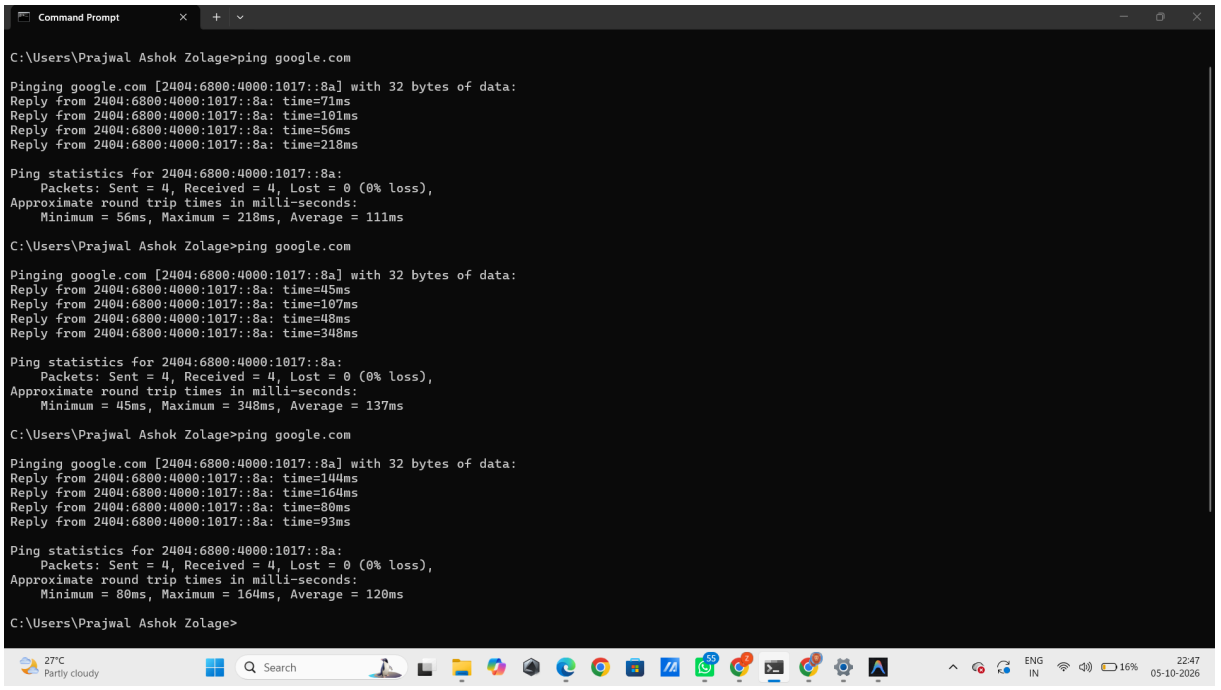
Result

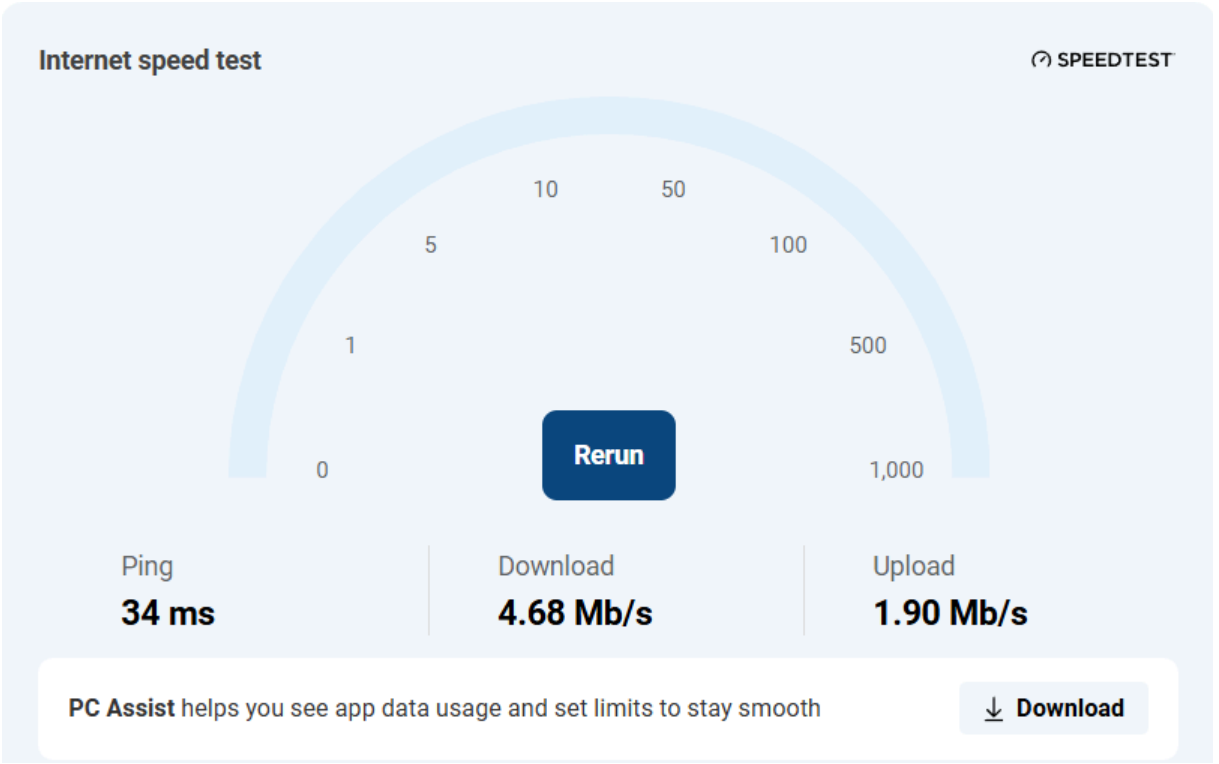
The **latency** and **throughput** of the network were measured successfully using Ping and Speed Test tools. The network performance can be evaluated based on **lower latency and higher throughput**.

Conclusion

Higher throughput + lower latency = better network performance.

Screenshot:


Command Prompt
C:\Users\Prajwal Ashok Zolage>ping google.com
Pinging google.com [2404:6800:4000:1017::8a] with 32 bytes of data:
Reply from 2404:6800:4000:1017::8a: time=71ms
Reply from 2404:6800:4000:1017::8a: time=101ms
Reply from 2404:6800:4000:1017::8a: time=56ms
Reply from 2404:6800:4000:1017::8a: time=218ms
Ping statistics for 2404:6800:4000:1017::8a:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 56ms, Maximum = 218ms, Average = 111ms
C:\Users\Prajwal Ashok Zolage>ping google.com
Pinging google.com [2404:6800:4000:1017::8a] with 32 bytes of data:
Reply from 2404:6800:4000:1017::8a: time=45ms
Reply from 2404:6800:4000:1017::8a: time=107ms
Reply from 2404:6800:4000:1017::8a: time=48ms
Reply from 2404:6800:4000:1017::8a: time=348ms
Ping statistics for 2404:6800:4000:1017::8a:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 45ms, Maximum = 348ms, Average = 137ms
C:\Users\Prajwal Ashok Zolage>ping google.com
Pinging google.com [2404:6800:4000:1017::8a] with 32 bytes of data:
Reply from 2404:6800:4000:1017::8a: time=144ms
Reply from 2404:6800:4000:1017::8a: time=164ms
Reply from 2404:6800:4000:1017::8a: time=80ms
Reply from 2404:6800:4000:1017::8a: time=93ms
Ping statistics for 2404:6800:4000:1017::8a:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 80ms, Maximum = 164ms, Average = 120ms
C:\Users\Prajwal Ashok Zolage>
27°C
Partly cloudy
Search
ENG
IN
16%
22:47
05-10-2026


Internet speed test
SPEEDTEST
0
1
5
10
50
100
500
1,000
Rerun
Ping
34 ms
Download
4.68 Mb/s
Upload
1.90 Mb/s
PC Assist helps you see app data usage and set limits to stay smooth
Download

Internet speed test

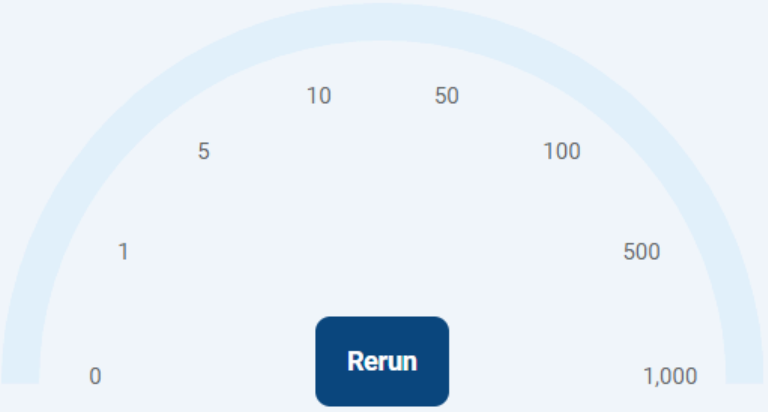
SPEEDTEST



Ping	Download	Upload
30 ms	**9.18 Mb/s**	**0.20 Mb/s**

Internet speed test

SPEEDTEST



Ping	Download	Upload
72 ms	**20.43 Mb/s**	**0.24 Mb/s**